Heat Transfer in TPMS-structures

Final Presentation

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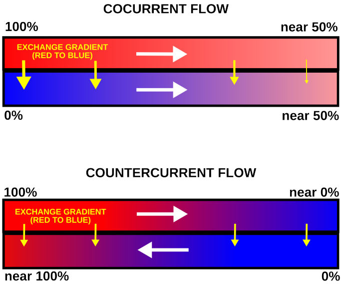
Heat Exchangers

Model the flow of heat by analysing:

- Fluid dynamics (Navier-Stokes eqns.)
- Heat conduction (Heat eqn.)



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- Heat exchangers are traditionally evaluated by use of a body-fitted mesh
- For complex geometries, this is unfeasible
- Warrants the investigation into alternative methods (Brinkman's method)
- Much of the theory is based on the paper [YJW+23] by Jiang et al.

Project Goal

The goal of this project is not to optimize a final industrial heat exchanger, but to test a modeling approach.

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- Can complex TPMS heat exchangers be simulated without body-fitted meshes?
- Can two independent fluid channels be represented in one computational domain?
- Can heat transfer be modeled using one global temperature field?
- We first use a simple sinusoidal wall as a proof-of-concept.

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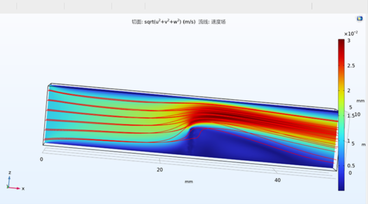
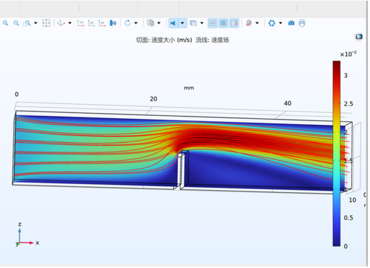
Fluid flow

Initially we implemented a simple model for fluid flow using the Brinkman method (left fig.), comparing it to the same geometry evaluated using traditional methods (right fig.).

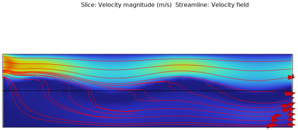


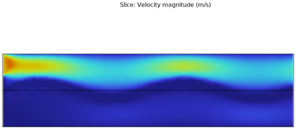


Note: All figures in this presentation are slices from 3D models.

Channel separation

Critical to the performance of the model is that the two channels are adequately separated, i.e. that there is no flow between the two fluids.
Simply implementing the Brinkman method 'as-is' does not separate the channels enough.





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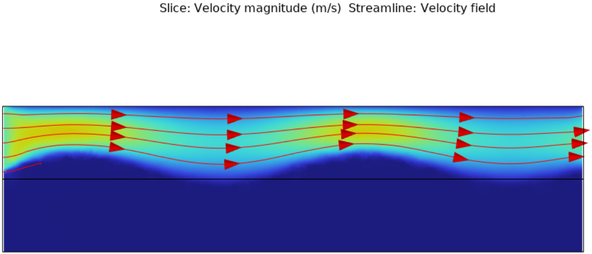
- Solid region: $|z - 0.05 - 0.01\sin(10\pi x)| < 0.005$
- Inlet velocity $1\,\mathrm{m/s}$
- Outlet pressure $0\,\mathrm{Pa}$
- Permeability 1E13 if channel else 1E-7

Single channel

For a single channel we can define *both* the wall and other channel so behave like solids, by changing the solid region to:

$$z - 0.05 - 0.01\sin(10\pi x) < 0.005.$$

This renders:



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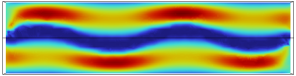
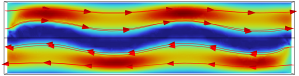
Dual channels

The Comsol framework can be adjusted to calculate the flow in two separate channels, by introducing an additional fluid domain:

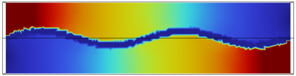
Slice: Velocity magnitude (m/s)



Slice: Velocity magnitude (m/s) Streamline: Velocity field



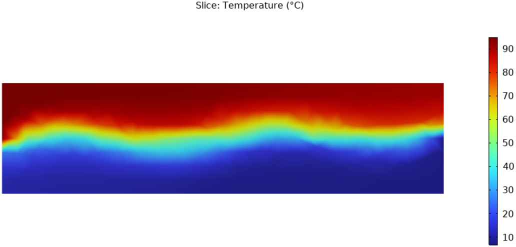
Slice: Pressure



$$z - 0.05 - 0.01\sin(10\pi x) < 0.005 \qquad \text{(Upper channel)}$$
$$z - 0.05 - 0.01\sin(10\pi x) > -0.005 \qquad \text{(Lower channel)}$$

Heat transfer

Using the same model as before we can turn on the *Heat Transfer in Fluids* module and solve for the temperature distribution. The thermal conductivity is set to $0.6\,\mathrm{W\,m^{-1}\,K^{-1}}$ for water and $400\,\mathrm{W\,m^{-1}\,K^{-1}}$ for copper:



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Background

Triply Periodic Minimal Surfaces (TPMS) incorporates a wide range of surfaces, of which the *gyroid* is the subject of this project. Discovered by NASA scientist Alan Schoen in 1970, it was proven to be embedded by Grosse-Brauckmann and Wohlgemuth in 1996 [Wik26].

- Embedded; divides space into two *disjoint* sub-volumes.

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- Can be approximated a the relatively simple equation

$$\sin x \cos y + \sin y \cos z + \sin z \cos x = 0$$

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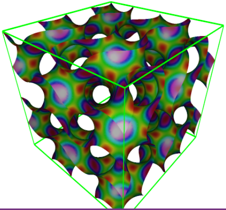

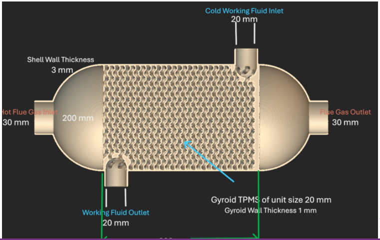

Advantages of TMPS

TPMS provides a compelling alternative to traditional fin-based heat exchangers.

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- High surface-to-volume ratio improves heat exchange.
- Continous TPMS walls generally provide good structural integrity (high stiffness).
- Geometry can be fine-tuned by function-based parameters.
- *Disadvantage:* Difficult to simulate performance

Modeling TPMS

The same method that was used for the simple sinusoidal geometry is applied to the TMPS structure.

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- The geometry is represented by continuous functions.
- Two Brinkman flow fields are solved, one for each fluid.
- A single temperature field is solved over the whole domain.
- Material properties are assigned using smooth interpolation functions.

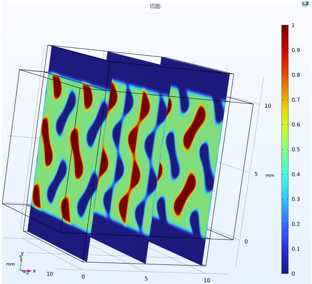
Material Fields

Instead of explicitly meshing the fluid-solid interfaces, the domain is divided by a smooth indicator function Φ.

$\Phi \approx 1$ fluid 1

$\Phi \approx 0.5$ solid wall

$\Phi \approx 0$ fluid 2



- The gyroid wall separates the two fluid channels.
- Inlet and outlet extensions are added to make boundary conditions easier.
- The smooth transition avoids discontinuous material properties.

Flow Model

Two Brinkman equations are used to model the two fluids separately.

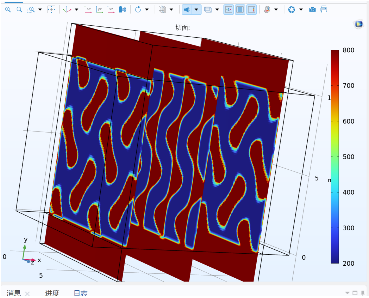
- For fluid 1, the fluid-1 region has low resistance.
- The wall and fluid-2 region are treated as high-resistance regions.
- Vice versa for fluid 2.

This allows the two flow fields to coexist in the same computational domain without explicitly meshing the interface.

Heat Transfer Model

The heat transfer problem is solved using one global temperature field; $\mathbf{u}_T = \mathbf{u}_{\text{fluid 1}} + \mathbf{u}_{\text{fluid 2}}$.

- Fluid regions: convection and conduction, Pe=800
- Wall region: conduction only, Pe=200
- The Péclet number (Pe) is interpolated between fluid and wall regions.



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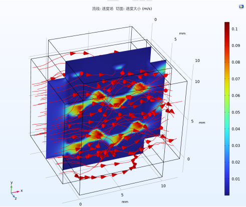
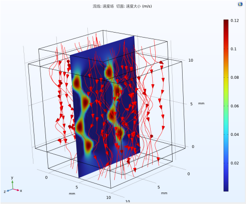
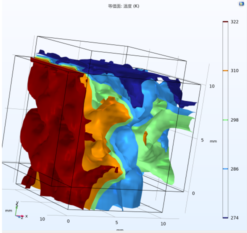
TPMS Simulation

The model was solved for a 50 K inlet temperature difference.

- The two velocity fields remain separated by the TPMS wall.
- The temperature field changes across the heat-exchanger core.
- Heat is transferred through the TPMS wall from the hot fluid to the cold fluid.

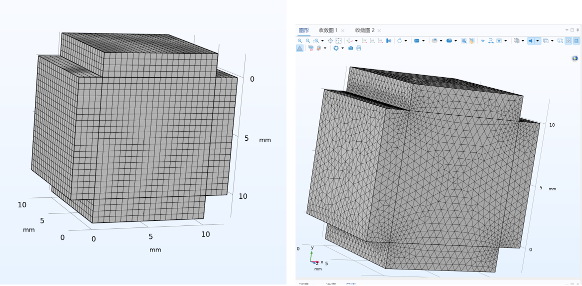






Discretizations

Two cases with different mesh discretizations were computed.



(a) Case 1 (b) Case 2

Figure 1: Different discretizations

Results

Parameter [Unit]	Case 1	Case2	Change
Hot-side inlet temp. [K]	323.15	323.15	
Hot-side avg. outlet temp. [K]	289.07	289.48	+0.41
Cold-side inlet temp. [K]	273.15	273.15	
Cold-side avg. outlet temp. [K]	293.61	295.88	+2.27
Hot-side pressure drop [Pa]	17.121	18.553	+8.36%
Cold-side pressure drop [Pa]	8.6676	9.4067	+8.53%

Conclusion

This method has a certain degree of robustness with respect to mesh discretization. Therefore, even if the local Φ boundary is rough, as long as the overall flow-channel volume fraction, resistance distribution, and heat-transfer area have similar averaged effects, the outlet temperature and pressure drop may still be close.

Discussion

The TPMS model demonstrates that the Brinkman-based method can be extended from a simple wall to a complex gyroid structure.

- Advantage: no body-fitted mesh is required for the internal TPMS wall.
- Advantage: geometry and wall thickness can be changed by parameters.
- Limitation: the result depends on mesh resolution and interpolation width.

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